

7.1.2 Software Requirements Analysis Process

The Software Requirements Analysis Process is a lower-level process of the Software Implementation Process. Users of ISO/IEC 15288 may choose to provide this process recursively using the Requirements Analysis Process of ISO/IEC 15288.

7.1.2.1 Purpose

The purpose of the Software Requirements Analysis Process is to establish the requirements of the software elements of the system.

7.1.2.2 Outcomes

As a result of successful implementation of the Software Requirements Analysis Process:

- a) Requirements for software elements and their interfaces are defined.
 - b) Software requirements are analyzed for correctness and testability.
 - c) The impact of software requirements on the operating environment is understood.
 - d) Consistency and traceability between software and system requirements are established.
 - e) Prioritization for implementing software requirements is determined.
 - f) Software requirements are approved and updated as needed.
 - g) Changes to software requirements are evaluated for cost, schedule, and technical impact.
 - h) Software requirements are baselined and communicated to all affected parties.
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7.1.2.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

7.1.2.3.1 Software Requirements Analysis

This activity defines and documents software requirements for each software item or configuration item.

- **7.1.2.3.1.1** Establish and document software requirements, including quality characteristics specifications.
- **7.1.2.3.1.2** Specify functional and capability requirements, performance requirements, physical characteristics, and environmental conditions for software item performance.
- **7.1.2.3.1.3** Define interfaces external to the software item.
- **7.1.2.3.1.4** Specify qualification requirements.
- **7.1.2.3.1.5** Specify safety requirements, including methods of operation, maintenance, environmental influences, and personnel injury prevention.
- **7.1.2.3.1.6** Specify security requirements, including prevention of compromise of sensitive information.
- **7.1.2.3.1.7** Specify human-factors engineering requirements, including manual operations, human–equipment interactions, personnel constraints, and areas sensitive to human error and training.
- **7.1.2.3.1.8** Specify data definition and database requirements.
- **7.1.2.3.1.9** Specify installation and acceptance requirements at the operation and maintenance sites.
- **7.1.2.3.1.10** Specify user documentation requirements.
- **7.1.2.3.1.11** Specify user operation and execution requirements.
- **7.1.2.3.1.12** Specify user maintenance requirements.
- **7.1.2.3.1.13** Determine and document the implementation priority of software requirements.
- **7.1.2.3.1.14** Record requirements and related artifacts in accordance with the **Software Documentation Management Process** (7.2.1) and place them under configuration control in accordance with the **Software Configuration Management Process** (7.2.2).

NOTE 1: Guidance for specifying quality characteristics is found in ISO/IEC 9126-1.

NOTE 2: If usability is an important requirement, recommendations for obtaining a desired level of usability are found in ISO/TR 18529, *Ergonomics of human-system interaction — Human-centred lifecycle process descriptions*. Annex E contains a process view focused on usability.

7.1.2.3.2 Requirements Evaluation and Approval

This activity evaluates the defined requirements, conducts reviews, and establishes baselines.

- **7.1.2.3.2.1** Evaluate software requirements using the following criteria and document the results:

- a) Traceability to system requirements and system design.
 - b) External consistency with system requirements.
 - c) Internal consistency.
 - d) Testability.
 - e) Feasibility of software design.
 - f) Feasibility of operation and maintenance.
 - **7.1.2.3.2.2** Conduct formal reviews of software requirements in accordance with the **Software Review Process** (7.2.6).
 - **7.1.2.3.2.3** Upon successful evaluation and review, approve and baseline software requirements.
 - **7.1.2.3.2.4** Communicate approved baselines to all affected parties.
 - **7.1.2.3.2.5** Evaluate subsequent changes to baselined software requirements for cost, schedule, and technical impact before approval.
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Cross-References

This process interacts with the following:

- **Software Documentation Management Process (7.2.1)** — for documenting requirements.
- **Software Configuration Management Process (7.2.2)** — for controlling requirement baselines.
- **Software Review Process (7.2.6)** — for conducting requirement reviews.
- **Software Architectural Design Process (7.1.3)** — for allocating software requirements to architecture.